Light

GCSE Computer Science - Legal, Ethical and Environmental Issues (Set 3)

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GCSE • OCR J277 • Grades 3—9 | Topic test | 30 min | 55 marks | Answered: 0 / 30

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Multiple-choice topic test for **GCSE Computer Science (OCR J277)**. You may skip questions. When you submit, you'll see your score, a grade estimate, and worked explanations. **Tip:** use the stopwatch above to time yourself.

Q1 (1 mark)

What does the accuracy principle of the Data Protection Act 2018 require?

- ○ **A.** Personal data must be accurate and, where necessary, kept up to date; inaccurate data should be corrected or deleted without delay
- ○ **B.** Data must be stored in alphabetical order
- ○ **C.** All data must be verified by a government official
- ○ **D.** Organisations must use accurate clocks on their computers

Q2 (1 mark)

What does Section 2 of the Computer Misuse Act 1990 specifically cover?

- ○ **A.** Installing antivirus software
- ○ **B.** Unauthorised access with intent to commit or facilitate further offences, such as fraud or blackmail
- ○ **C.** Using social media during work hours
- ○ **D.** Forgetting your own password

Q3 (1 mark)

What are moral rights in copyright law?

- ○ **A.** Personal rights of creators including the right to be identified as the author and the right to object to derogatory treatment of their work
- ○ **B.** The right to charge money for creative works
- ○ **C.** The right to copy any work for personal use
- ○ **D.** Rights that only apply to religious content

Q4 (1 mark)

What characterises the MIT software licence?

- ○ **A.** Software can only be used at the Massachusetts Institute of Technology
- ○ **B.** All software must be kept secret and never shared
- ○ **C.** A permissive licence allowing free use, modification, and distribution with minimal restrictions - only requiring the licence notice to be included

- ○ **D.** Users must pay annual fees to use the software

Q5 (1 mark)

What are deepfakes?

- ○ **A.** Very deep swimming pools used in technology testing
- ○ **B.** Websites that sell counterfeit products
- ○ **C.** A type of secure encryption
- ○ **D.** AI-generated synthetic media that can create realistic fake videos, images, or audio of people saying or doing things they never did

Q6 (1 mark)

Why is cryptocurrency mining considered an environmental concern?

- ○ **A.** Some cryptocurrencies require enormous computing power for mining, consuming vast amounts of electricity and contributing significantly to carbon emissions
- ○ **B.** Mining physically damages the ground
- ○ **C.** Cryptocurrency miners work in dangerous underground conditions
- ○ **D.** Digital coins are made from rare metals

Q7 (1 mark)

What does the term 'digital native' mean?

- ○ **A.** A person who was born in a country with fast internet
- ○ **B.** Someone who grew up with digital technology from childhood and is typically comfortable using it intuitively
- ○ **C.** A computer program written in a native programming language
- ○ **D.** A person who refuses to use any technology

Q8 (1 mark)

What is biometric data?

- ○ **A.** Data about biological research experiments
- ○ **B.** Information about a person's biography
- ○ **C.** Personal data derived from physical or behavioural characteristics that can uniquely identify a person, such as fingerprints, facial features, or voice patterns
- ○ **D.** Medical prescriptions stored electronically

Q9 (1 mark)

What does the storage limitation principle of the Data Protection Act 2018 require?

- ○ **A.** Data must be stored in the smallest possible file format
- ○ **B.** Only 1GB of data can be stored per person
- ○ **C.** Data must be stored on servers located in the UK only

○ **D.** Personal data should not be kept longer than necessary for the purposes for which it was collected

Q10 (1 mark)

Why is digital addiction considered an ethical concern for technology companies?

○ **A.** Companies deliberately design apps with features like infinite scroll and notifications to maximise engagement, potentially harming users' mental health and wellbeing

○ **B.** Addicted users buy fewer products

○ **C.** Digital addiction only affects elderly users

○ **D.** Technology cannot be addictive

Q11 (1 mark)

Why is recycling rare earth elements from electronic devices important?

○ **A.** Rare earth elements are not actually used in electronics

○ **B.** Rare earth elements are only found on the moon

○ **C.** Mining rare earths causes significant environmental damage; recycling recovers valuable materials while reducing mining demand and preventing toxic e-waste contamination

○ **D.** Recycled materials cannot be reused in new products

Q12 (1 mark)

What is the main difference between copyright and patents?

- ○ **A.** Copyright protects creative expression like writing and art; patents protect inventions and technical processes
- ○ **B.** Copyright is free but patents require payment
- ○ **C.** Patents last forever but copyright expires
- ○ **D.** There is no difference - they are the same thing

Q13 (2 marks)

A company uses a third-party cloud service to store customer data. Under the Data Protection Act 2018, what are the responsibilities of each party?

- ○ **A.** The cloud service is solely responsible for all data protection compliance
- ○ **B.** The company remains the data controller responsible for compliance; the cloud service is a data processor who must follow the controller's instructions and maintain appropriate security
- ○ **C.** Neither party has any legal responsibilities for the data
- ○ **D.** Customers are responsible for protecting their own data once submitted

Q14 (2 marks)

A hacker is convicted under Section 3 of the Computer Misuse Act for deploying ransomware that affected hospital systems.
What factors might influence the severity of their sentence?

- ○ **A.** Sentences are always fixed regardless of circumstances

- **B.** Only the hacker's age affects sentencing
- **C.** Computer crimes always receive community service only
- **D.** Severity depends on factors including the scale of damage, whether critical infrastructure was affected, financial losses caused, whether lives were endangered, and whether the offender profited from the crime

Q15 (2 marks)

Digital Rights Management (DRM) systems are used by content providers to control how digital content can be used.
What issues does DRM raise?

- **A.** DRM has no drawbacks and benefits everyone equally
- **B.** DRM is illegal in all countries
- **C.** DRM can prevent legitimate uses like backups, accessibility features, and format shifting; it may stop working if companies cease support; circumventing DRM may be illegal even for legitimate purposes
- **D.** DRM only applies to physical products like DVDs

Q16 (2 marks)

Some software is released under 'dual licensing' - available under both an open source licence and a commercial licence.
What is the purpose of this approach?

- **A.** It allows free use under open source terms for community and non-commercial purposes, while generating revenue from businesses who need commercial licence terms such as proprietary integration, indemnification, or support

- **B.** Dual licensing has no practical purpose
- **C.** Users must purchase both licences to use the software
- **D.** Dual licensing means the software has two different names

Q17 (2 marks)

Technology platforms enable the 'gig economy' where workers complete short-term tasks rather than traditional employment.
What ethical concerns does this raise?

- **A.** The gig economy has no ethical concerns
- **B.** Only older workers participate in the gig economy
- **C.** Gig economy workers always earn more than traditional employees
- **D.** Workers may lack employment protections like minimum wage, sick pay, and pensions; algorithmic management reduces human oversight; income instability affects worker wellbeing; platforms profit while shifting risks to workers

Q18 (2 marks)

Video streaming services have become a major source of internet traffic. What is the environmental impact of streaming?

- **A.** Streaming has no environmental impact because it is digital
- **B.** Streaming requires data centres, network infrastructure, and user devices all consuming electricity; high-definition and 4K content increases data transfer and energy use; the growing popularity of streaming significantly contributes to technology's carbon footprint
- **C.** Streaming reduces environmental impact compared to all

alternatives

○ **D.** Only downloading videos uses energy; streaming uses none

Q19 (2 marks)

Some countries implement internet censorship, blocking access to certain websites and content.
What are arguments for and against this practice?

○ **A.** Internet censorship is universally supported by all governments

○ **B.** Internet censorship is technically impossible to implement

○ **C.** Proponents argue it protects citizens from harmful content, maintains social stability, and protects national security; opponents argue it violates freedom of expression, enables authoritarian control, blocks legitimate information, and is often applied inconsistently

○ **D.** Only small countries practice internet censorship

Q20 (2 marks)

Data brokers collect and sell personal information about individuals. What privacy concerns do they raise?

○ **A.** Individuals often do not know brokers hold their data; information is aggregated from many sources creating detailed profiles; data may be inaccurate but still affect decisions; individuals have limited ability to access, correct, or delete this data

○ **B.** Data brokers have no impact on privacy

○ **C.** Data brokers only collect publicly available information

○ **D.** Data brokers must get explicit consent before collecting any data

Q21 (2 marks)

Algorithmic decision-making affects many aspects of life, from loan applications to content recommendations.
Why is algorithmic transparency an ethical concern?

○ **A.** Algorithms always make perfectly fair decisions

○ **B.** Transparency would make algorithms work better technically

○ **C.** Only computer scientists are affected by algorithmic decisions

○ **D.** People affected by algorithmic decisions often cannot understand or challenge how decisions are made; algorithms may contain hidden biases; lack of transparency prevents accountability; automated decisions may lack human oversight for edge cases

Q22 (2 marks)

Software as a Service (SaaS) delivers applications over the internet rather than through local installation.
How does this affect software licensing?

○ **A.** SaaS has no licensing implications

○ **B.** Users subscribe rather than own software; access can be terminated by the provider; data portability becomes a concern; users depend on continued provider support; licensing shifts from one-time purchase to ongoing subscription

○ **C.** SaaS software cannot be licensed

○ **D.** Users own the software permanently with SaaS

Q23 (2 marks)

Automated profiling uses algorithms to evaluate aspects of a person's life, such as predicting job performance or creditworthiness.
What rights do individuals have regarding profiling under data protection law?

- ○ **A.** Individuals have no rights regarding automated profiling
- ○ **B.** Only businesses can be profiled, not individuals
- ○ **C.** Individuals have the right to be informed about profiling, to object to it in certain circumstances, and to not be subject to decisions based solely on automated processing that significantly affect them without human involvement
- ○ **D.** Profiling is illegal in all circumstances

Q24 (3 marks)

Deepfake technology can create realistic fake videos of politicians, celebrities, and ordinary people.
Analyse the ethical, legal, and societal implications.

- ○ **A.** Deepfakes are harmless entertainment with no negative implications
- ○ **B.** Ethically: violates consent and can destroy reputations; creates non-consensual intimate imagery; enables fraud and manipulation. Legally: existing laws struggle to address novel harms; defamation and fraud laws may apply but are inadequate. Societally: erodes trust in video evidence; enables political disinformation; creates 'liar's dividend' where real footage can be dismissed as fake
- ○ **C.** Deepfakes only affect famous people and have no broader impact
- ○ **D.** Technology will automatically detect all deepfakes, eliminating concerns

Q25 (3 marks)

A company claims their cloud service is 'carbon neutral' because they purchase carbon offsets.
Analyse the validity of this environmental claim.

- ○ **A.** Carbon offset purchases make any activity genuinely carbon neutral
- ○ **B.** Carbon offsets are illegal and should not be used
- ○ **C.** Cloud services have no carbon emissions to offset
- ○ **D.** While offsets can contribute to environmental efforts, they do not eliminate emissions at source; offset quality varies greatly; some offset projects fail to deliver promised reductions; 'carbon neutral' claims may mislead consumers about actual environmental impact; reducing emissions directly is more effective than offsetting

Q26 (3 marks)

A company plans to implement facial recognition for employee access control.
Analyse the Data Protection Act requirements before proceeding.

- ○ **A.** No special requirements apply because employees can be required to comply with workplace rules
- ○ **B.** A Data Protection Impact Assessment (DPIA) is likely required because biometric data is special category data; the assessment must evaluate necessity and proportionality; less intrusive alternatives must be considered; lawful basis for processing must be established; employees cannot be forced to consent as consent must be freely given
- ○ **C.** Facial recognition is automatically compliant with data protection law
- ○ **D.** Only customer-facing facial recognition requires assessment

Q27 (3 marks)

A security researcher discovers a vulnerability in a major company's website. They access the system to confirm the vulnerability exists, document it, and report it to the company.
Analyse the legal and ethical position.

- ○ **A.** This is always completely legal because they intended to help
- ○ **B.** Security researchers are completely protected from prosecution
- ○ **C.** Only the company can decide whether a law was broken
- ○ **D.** Legally complex: accessing without authorisation likely violates Section 1 of the Computer Misuse Act regardless of intent; 'good faith' security research lacks explicit legal protection in UK law; ethically, responsible disclosure serves public interest; companies may choose not to prosecute helpful researchers but legally could do so; the law does not currently distinguish malicious hackers from security researchers

Q28 (3 marks)

Virtual and augmented reality technologies are becoming increasingly immersive and widely adopted.
Analyse the cultural and societal implications of widespread VR/AR adoption.

- ○ **A.** VR/AR will have no significant cultural impact
- ○ **B.** VR/AR technology is too expensive to ever become widespread
- ○ **C.** Positive: new forms of creative expression, education, and connection; accessibility for those with physical limitations; immersive cultural experiences. Concerns: further blurring of physical and digital life; potential for increased isolation; questions about virtual property and identity; new forms of harassment in virtual spaces; unknown psychological effects of extended immersion

○ **D.** Only gaming will be affected by VR/AR technology

Q29 (3 marks)

A company discovers that code they open-sourced under a permissive licence (MIT) is being used by a competitor in a proprietary product. Analyse the legal and strategic implications.

○ **A.** The competitor has violated copyright by using the code

○ **B.** Legally, the competitor is within their rights - permissive licences explicitly allow commercial and proprietary use without requiring disclosure; strategically, this demonstrates both the benefits (widespread adoption) and drawbacks (enabling competitors) of permissive licensing; the company cannot retrospectively change licence terms for already-released code

○ **C.** Open source code can never be used in proprietary products

○ **D.** The company can sue to recover their code

Q30 (3 marks)

Health tracking apps collect sensitive data about users' exercise, heart rate, sleep patterns, and sometimes menstrual cycles or mental health. Analyse the privacy implications of sharing this data with app providers.

○ **A.** Health data from apps is not sensitive and requires no special protection

○ **B.** Health apps never share data with third parties

○ **C.** Users have complete control over all health data collected by apps

○ **D.** Health data is sensitive under data protection law requiring higher protections; apps may share data with advertisers, insurers, or employers with serious consequences; aggregated health data creates

detailed profiles; security breaches expose highly personal information; users often grant permissions without understanding implications; data may be used in ways that disadvantage users

Total: **55** marks • Time: **30 min**

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